

3D open source robots soon to become a reality

If you are a fan of sci-fi movies and yearn for a day when cyborgs rule the roost, then there's good news for you. According to Intel, the international tech giant, 3D open source robots will soon be among us to bridge the gap between technology and humans.

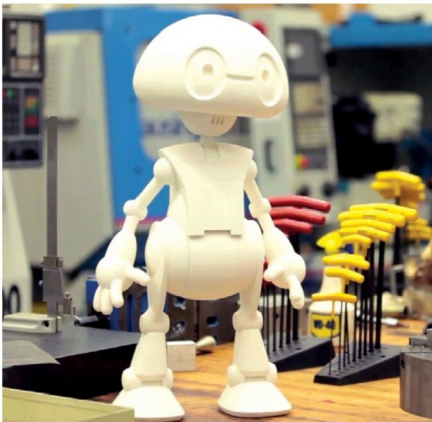
The announcement was made by Brian David Johnson at the Intel Developer Forum (IDF) in San Francisco. "We've been doing this thing called science fiction prototyping... something that you haven't known about, using science fiction inside of Intel," said Johnson. "The idea is that there is no line between technology and humanity: we and you — our technology with our humanity — [robots] that share our sense of culture, morals, with even our dreams of the future," Johnson fantasised.

"So as we begin to make these robots and design and build them, we begin to see these robots as an extension of ourselves, and allow them to go off and interact with other people, [and] with other robots," he added.

The project, which may sound a little crazy, is basically trying to create robots, and change and share them in online communities, enabling the various designs to be printed in 3D. According to Johnson, Intel had already started the process of collaborating with researchers all over the world around ten years ago, as part of the research project. "Jimmy was our first example, a real robot with real AI," Johnson said.

Robots may replace manual labour soon

With increasing wages and an aging population, it seems like robotic workers will replace people on factory floors in the near future. China will soon have industrial robots, ranging from high-end humanoid machines that can



Intel's 3D open source robot Jimmy

see, touch and even learn, to low-cost robots that compete with unskilled labour to undercut minimum wages in the country.

In the next five years, China's factories will have a huge shortfall of labour as the country's youth is unwilling to perform manual labour.

It is not just the traditional robot manufacturers such as Zurich-based ABB Group and Germany's Kuka that are active in this domain. Even large Asian electronics suppliers like Delta Electronics Inc. and Foxconn Technology Group are looking forward to constructing robots, along with smaller players such as Universal Robots A/S from Denmark.

However, some industry executives warn that China's automation switch may take years as there are several challenges, such as the high

cost of advanced robots, increasing technical limitations and even the lack of flexibility associated with bringing robots into the factory. Said Tim Li, senior vice president of Taiwanese PC contract manufacturer, Quanta Computer Inc., "If your orders decrease, you can lay off workers. But you can't lay off robots."

Charge smartphones using microwaves

It may sound strange but a microwave oven could very soon be able to charge your mobile phone. According to recent reports, researchers at the University of Tokyo have been working on ways to do this, using the energy lost by microwaves while heating to charge mobile phones.

The team, headed by engineer Yoshihiro Kawahara, is developing a